

Supplementary Material for “Locality Sensitive Hashing-based Knowledge Transfer for Multitask Optimization”

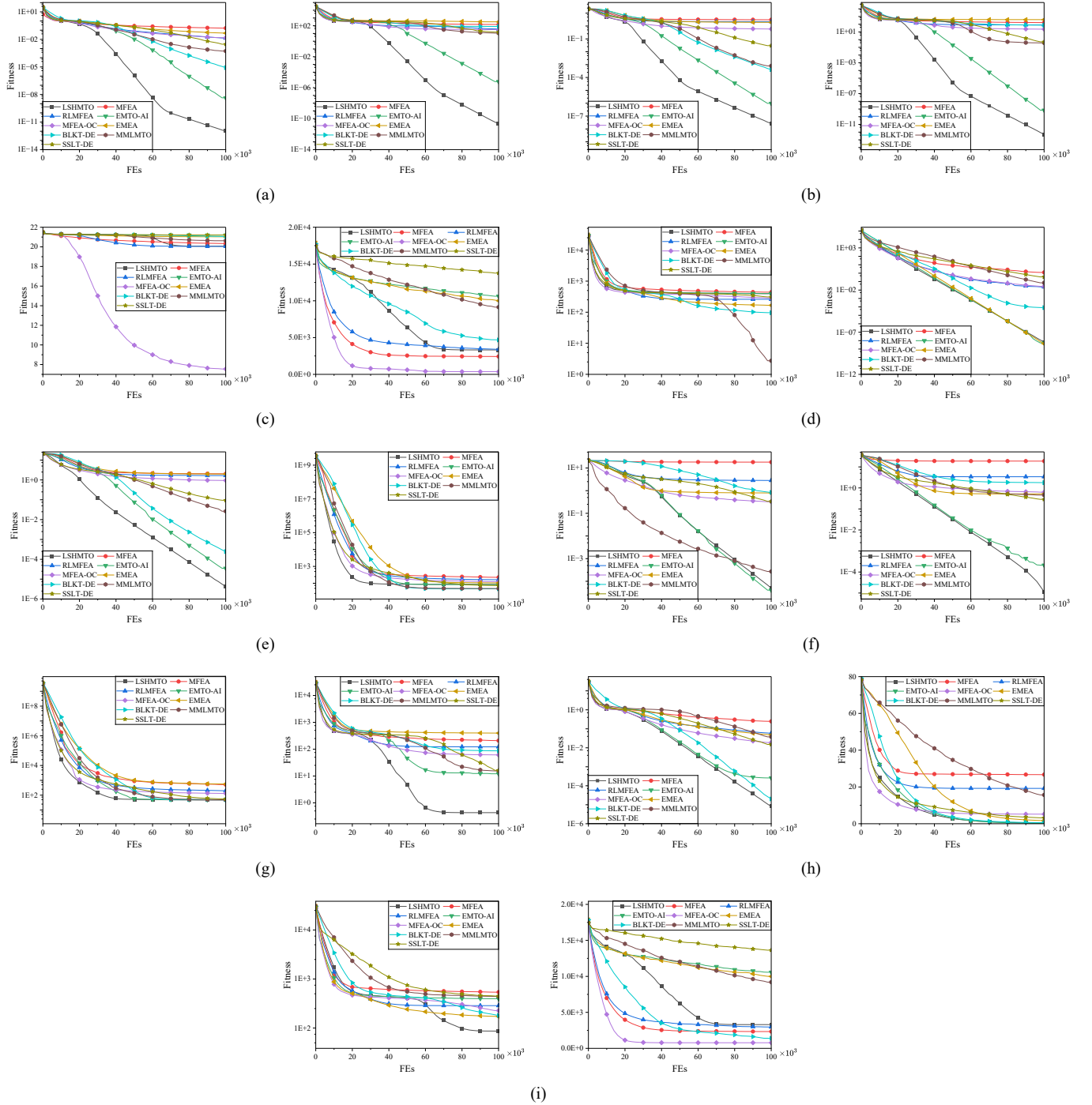


Fig. S1. Convergence curves for LSHMTO and compared EMTO algorithms on CEC17. (a) CIHS; (b) CIMS; (c) CILS; (d) PIHS; (e) PIMS; (f) PILS; (g) NIHS; (h) NIMS; (i) NILS.

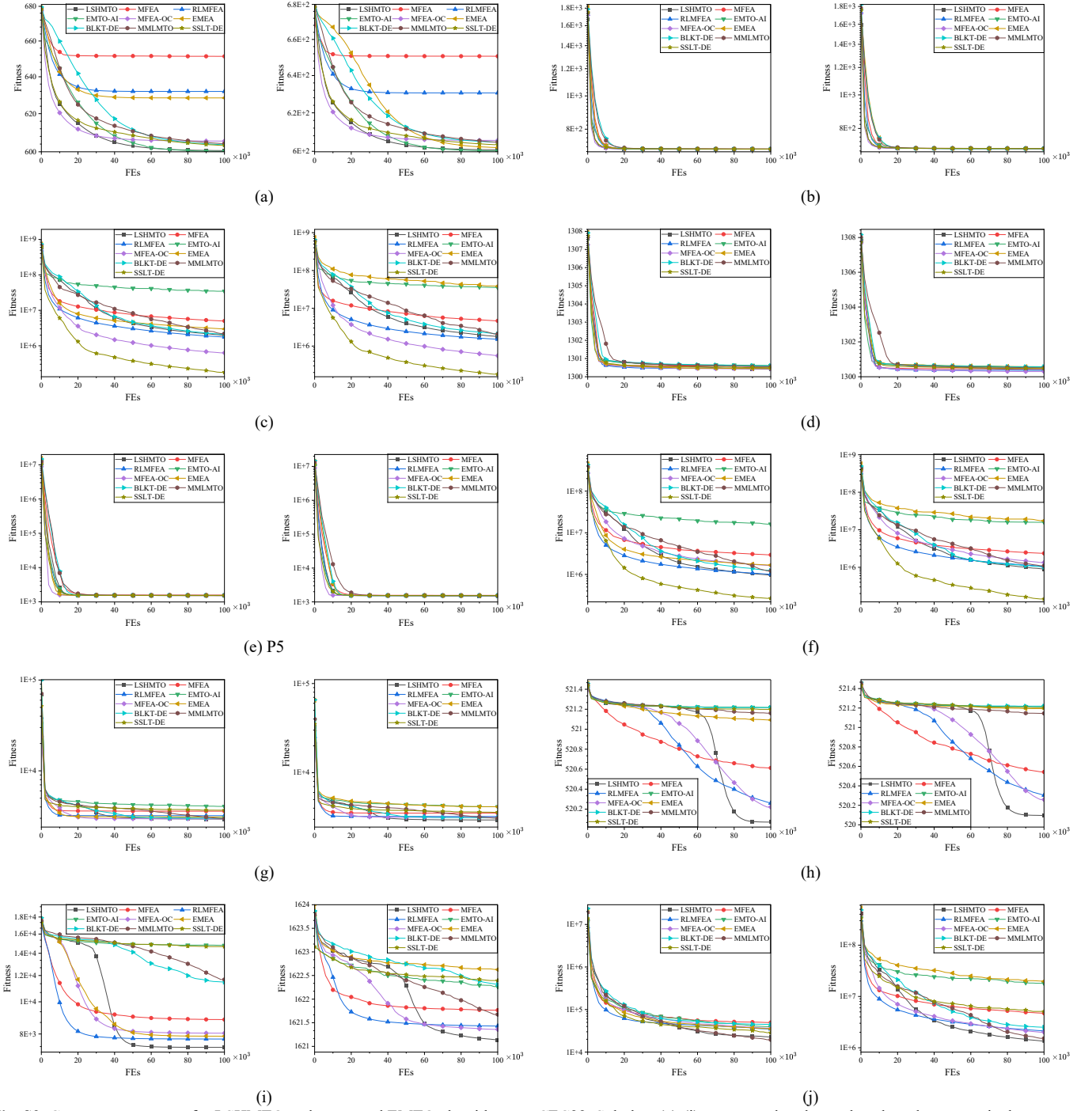


Fig. S2. Convergence curves for LSHMTO and compared EMTO algorithms on CEC22. Subplots (a)-(j) correspond to the ten benchmarks, respectively.

TABLE S.I
THE COMPLETE COMPARISON RESULTS AMONG LSHMTO WITH DIFFERENT ktp VALUE

Problem		LSHMTO($ktp=0.1$)	$ktp=0.0$	$ktp=0.2$	$ktp=0.3$	$ktp=0.5$	$ktp=0.6$	$ktp=0.7$	$ktp=0.9$	$ktp=1.0$
CIHS	T1	1.0353e-12	8.5012e-6(+)	2.5150e-4(+)	1.8741e-3(+)	4.7665e-3(+)	3.7390e-3(+)	4.3049e-3(+)	4.2117e-3(+)	4.1252e-2(+)
	T2	2.1212e-11	3.9580e+2(+)	2.2232e+0(+)	1.0117e+1(+)	3.2276e+1(+)	5.1584e+1(+)	1.2087e+2(+)	1.4967e+2(+)	4.9848e+1(+)
CIMS	T1	2.6481e-8	1.4600e+0(+)	2.5730e-3(+)	7.2994e-3(+)	4.3051e-1(+)	1.3952e+0(+)	3.4179e+0(+)	7.4383e+0(+)	2.9534e+0(+)
	T2	5.1728e-13	3.9805e+2(+)	2.9168e-2(+)	4.1696e-1(+)	2.5842e+1(+)	5.0551e+1(+)	1.8497e+2(+)	3.5622e+2(+)	1.1613e+2(+)
CILS	T1	2.0066e+1	2.1194e+1(+)	2.0112e+1(≈)	2.0152e+1(≈)	2.0248e+1(≈)	2.0147e+1(-)	2.0227e+1(-)	2.0127e+1(-)	2.0041e+1(-)
	T2	3.3215e+3	1.0078e+4(+)	3.8471e+3(+)	4.0006e+3(+)	4.6253e+3(+)	5.0357e+3(+)	5.8109e+3(+)	7.5324e+3(+)	9.0331e+3(+)
PIHS	T1	3.9418e+2	4.0211e+2(+)	1.1495e+2(-)	9.6954e+1(-)	1.1841e+2(-)	1.2992e+2(-)	1.7579e+2(-)	2.9248e+2(-)	3.9822e+2(≈)
	T2	6.8767e-9	3.0563e-9(-)	1.9817e-4(-)	1.1785e-3(+)	1.8327e-3(+)	5.8164e-4(+)	5.4771e-4(+)	6.0327e-4(+)	3.9991e-3(+)
PIMS	T1	4.2877e-6	1.1281e+0(+)	3.4242e-2(≈)	6.6794e-1(+)	1.8456e+0(+)	1.8894e+0(+)	1.8809e+0(+)	1.2931e+0(+)	9.4684e-1(+)
	T2	8.5704e+1	6.1474e+1(-)	8.4021e+1(≈)	9.6297e+1(+)	1.1584e+2(+)	1.1786e+2(+)	1.1725e+2(+)	1.1470e+2(+)	1.1989e+2(+)
PILS	T1	5.0373e-5	4.4659e+0(+)	1.8480e-5(-)	2.9597e-5(-)	4.5912e-4(+)	3.9578e-1(+)	9.4574e-1(+)	6.5998e+0(+)	3.1582e+0(+)
	T2	1.1454e-5	6.3827e-1(+)	1.4455e-5(≈)	1.5817e-1(+)	3.9699e-1(+)	5.0460e-1(+)	6.7653e-1(+)	4.7573e+0(+)	3.4333e+0(+)
NIHS	T1	4.6739e+1	7.1177e+1(+)	6.1462e+1(≈)	7.6433e+1(≈)	1.2636e+2(+)	2.2465e+2(+)	1.3085e+2(+)	9.3582e+1(+)	1.9170e+2(+)
	T2	4.3115e-1	3.9679e+2(+)	9.7853e+0(+)	2.0300e+1(+)	4.3783e+1(+)	6.9689e+1(+)	1.2073e+2(+)	2.1391e+2(+)	7.2298e+1(+)
NIMS	T1	8.1544e-6	7.9488e-6(≈)	1.9563e-4(+)	6.3446e-4(+)	7.5124e-4(+)	3.4265e-3(+)	3.0107e-3(+)	6.4635e-3(+)	2.4187e-2(+)
	T2	4.0389e-1	2.0539e+0(+)	2.5307e+0(+)	3.8923e+0(+)	8.8013e+0(+)	1.1833e+1(+)	1.4535e+1(+)	2.0821e+1(+)	1.9658e+1(+)
NILS	T1	8.6808e+1	3.9434e+2(+)	8.4307e+1(≈)	8.9053e+1(≈)	1.0085e+2(≈)	1.0597e+2(+)	1.1682e+2(+)	1.9204e+2(+)	3.6929e+2(+)
	T2	3.2994e+3	1.0253e+4(+)	4.0764e+3(+)	4.1656e+3(+)	4.6726e+3(+)	5.0725e+3(+)	5.7386e+3(+)	7.3078e+3(+)	8.4054e+3(+)
Number of + / ≈ / -			15 / 1 / 2	9 / 6 / 3	13 / 3 / 2	15 / 2 / 1	16 / 0 / 2	16 / 0 / 2	16 / 0 / 2	16 / 1 / 1